

1984. Minimum Difference Between Highest and Lowest of K Scores

Solved 

Easy

 Topics

 Companies

 Hint

You are given a **0-indexed** integer array `nums`, where `nums[i]` represents the score of the i^{th} student. You are also given an integer `k`.

Pick the scores of any `k` students from the array so that the **difference** between the **highest** and the **lowest** of the `k` scores is **minimized**.

Return *the **minimum** possible difference*.

Example 1:

Input: `nums = [90]`, `k = 1`

Output: `0`

Explanation: There is one way to pick score(s) of one student:

- `[90]`. The difference between the highest and lowest score is $90 - 90 = 0$.

The minimum possible difference is 0.

Example 2:

Input: `nums = [9,4,1,7]`, `k = 2`

Output: `2`

Explanation: There are six ways to pick score(s) of two students:

- `[9,4,1,7]`. The difference between the highest and lowest score is $9 - 4 = 5$.

- `[9,4,1,7]`. The difference between the highest and lowest score is $9 - 1 = 8$.

- `[9,4,1,7]`. The difference between the highest and lowest score is $9 - 7 = 2$.

- `[9,4,1,7]`. The difference between the highest and lowest score is $4 - 1 = 3$.

- `[9,4,1,7]`. The difference between the highest and lowest score is $7 - 4 = 3$.

- `[9,4,1,7]`. The difference between the highest and lowest score is $7 - 1 = 6$.

The minimum possible difference is 2.

Constraints:

- `1 <= k <= nums.length <= 1000`
- `0 <= nums[i] <= 105`

Python:

class Solution:

```
def minimumDifference(self, nums: List[int], k: int) -> int:  
    return nums.sort() or min(R-L for L, R in zip(nums, nums[k-1:]))
```

JavaScript:

```
var minimumDifference = function(nums, k) {  
    if (k === 1) return 0;  
  
    nums.sort((a, b) => a - b);  
    let minDiff = Infinity;  
  
    for (let i = 0; i <= nums.length - k; i++) {  
        minDiff = Math.min(minDiff, nums[i + k - 1] - nums[i]);  
    }  
  
    return minDiff;  
};
```

Java:

```
class Solution {  
    public int minimumDifference(int[] nums, int k) {  
        int n = nums.length;  
        Arrays.sort(nums);  
        int ans = nums[k - 1] - nums[0];  
        for (int i = 0; i + k <= n; i++) {  
            ans = Math.min(ans, nums[i + k - 1] - nums[i]);  
        }  
        return ans;  
    }  
}
```